

Workshop 3: Analysing Job Market Data With LLM APIs

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Introduction:

This project analyses data from Data Analyst job postings, including salary estimates, minimum experience requirements and preferred data languages. The aim is to explore how experience and programming language requirements relate to salary outcomes in the job market.

Figure 1: Years of Experience Analysis

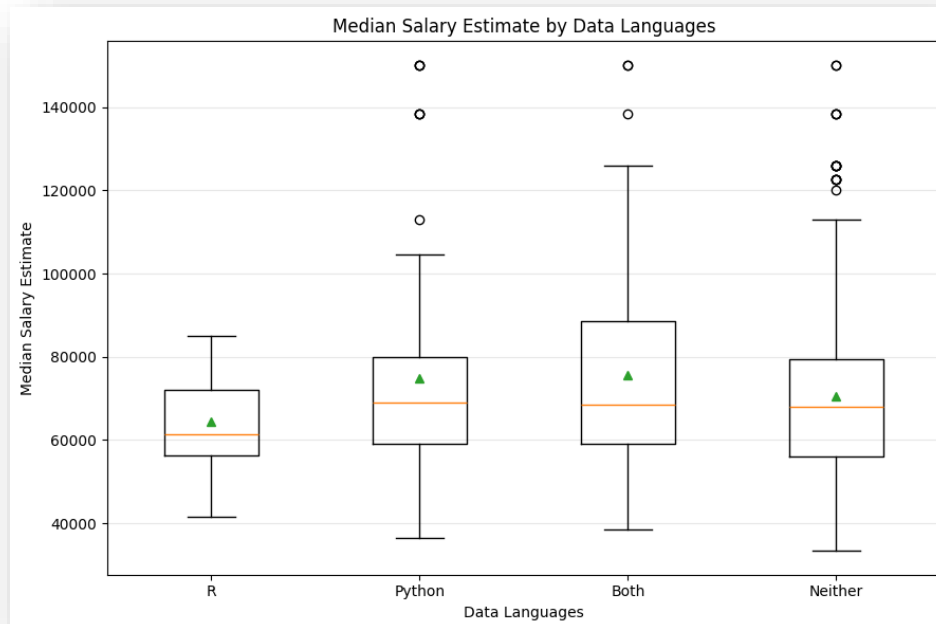


There is no strict linear relationship between minimum years of experience and median salary. High salaries appear at multiple experience levels, indicating that experience alone does not fully explain the salary variation.

Interpretation:

- The scatter points are widely dispersed, showing substantial salary variation at nearly every experience level.
- While some higher salaries cluster around mid-career experience (e.g., 4–8 years), there are also high-salary points at low experience levels, suggesting that other things like role type, specialisation, or industry may matter more than tenure alone.
- The absence of a clear upward trend indicates that experience is not a strong standalone predictor of salary in this dataset. Or it could simply reflect the absence of insufficient data and not a lack of relationship.
- The spread of points at each experience level suggests that other factors including technical skills, job function, sector, or location etc., likely drive salary differences more strongly than years of experience. Or because of the limited data set, there is no indicative trend even if a real trend did exist.
- Overall, the figure highlights a complex, non-linear relationship. While experience contributes to salary it does not determine it. But one cannot conclude that experience has no impact. The dataset is too small to rule out a trend.

Figure 2: Programming Language Analysis:



In terms of the highest median salary estimates, the people who use both R and Python tend to have the highest median salary estimates, while those who use neither have the lowest. The salary distribution also varies substantially by language group.

Interpretation:

- The 'Both' category shows the highest median and mean salary levels, suggesting that those proficient in both R and Python skills have stronger earning potential. This may reflect higher technical versatility or roles requiring broader analytical capability. Because of the small data set in each language category, the median and mean can become unstable too and one unusually high or low salary can shift the whole box.
- Python-only users appear to earn more than R-only users on average, though the difference is not extreme.
- The 'Neither' group has the lowest median salary and the narrowest interquartile range, indicating both lower pay and less variation in roles.
- The presence of outliers above the boxes especially in the Python and 'Both' groups might suggest that some individuals in these categories command significantly higher salaries, maybe due to seniority or specialised roles. Or they might simply reflect the randomness of a smaller data set.
- Overall, the pattern implies a positive association between programming language proficiency and salary, with combined R and Python skills offering the strongest advantage. In saying that, while we can still describe patterns, the trends must be treated as suggestive and not conclusive. The graph shows possible trends only.

Reflection:

This was a useful exercise for me in understanding how AI reporting tools can work with a dataset to generate meaningful visualisations which we can then interpret and also seeing how quickly visualisations can be generated, surfacing patterns that are not immediately obvious from raw tables. At the same time, the exercise also reinforced the importance of critical thinking when working with

small data sets. Reflecting on this limitation was an important reminder that visualisations are not truth machines but are tools that can help us explore data, and that they must be interpreted with caution and contextual awareness.

Another key learning for me was the value of clear, intentional prompting. By specifying the exact columns and the type of plots needed, the system was able to generate the visuals efficiently. Again, this reiterates the importance of articulating analytical goals precisely and is a skill that translates directly to how we work with real-world data, where the clarity of intent often determines the quality of the output.

Overall, this exercise opened my eyes up to the power of AI as reporting tools and greatly enhanced my understanding of how to use AI and move from raw data to visual insights with it.